

Muhammad Inshal

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SUMMARY

I am motivated Computer Science student at FAST-NUCES with a strong foundation in programming and a passion for building practical solutions. Continuously improving problem-solving skills through LeetCode. Actively share coding resources and useful notes on GitHub, including JavaScript, HTML, and Python content. Eager to contribute to meaningful projects and turn ideas into real-world solutions.

EDUCATION

FAST-NUCES, Faisalabad

07/2022 - 06/2026

Bachelor of Science in Computer Science

Coursework: OOP, Data Structures & Algorithms, Database, Operating Systems, Computer Networks, AI, GenAI

ACHIEVEMENTS

Solved 250+ Data Structures and Algorithms problems on LeetCode, demonstrating consistent problem-solving and logical thinking skills.

CERTIFICATIONS

Docker for Absolute Beginners — *DataCamp*

The Git & GitHub Bootcamp — *Udemy*

PROJECTS

TestHub — *React, TypeScript, Go, MongoDB, Docker, FastAPI, Gemini* 

- Built a web-based automated testing platform with Selenium and Playwright support, parallel execution via Selenium Grid and Docker, and a centralized dashboard for results, logs, screenshots, videos, and reports.
- Built an AI-powered failure diagnosis module using FastAPI and Google Gemini to analyze logs, stack traces, and screenshots, delivering root-cause analysis and debugging recommendations.
- Containerized all services with Docker and built a GitHub Actions CI/CD pipeline for automated builds, testing, deployment, and downloadable reports and artifacts.

Neural Storyteller: Image Captioning — *PyTorch, torchvision* 

- Built an end-to-end image captioning pipeline using ResNet50 for feature extraction and an LSTM-based Seq2Seq decoder.
- Implemented greedy decoding and beam search strategies to improve caption quality, with a Gradio web app for real-time inference.

Masked Autoencoder (MAE) — *PyTorch, Vision Transformer* 

- Designed a self-supervised MAE using an asymmetric ViT encoder-decoder: ViT-Base (768-dim) encoding only 25% visible patches and a lightweight ViT-Small (384-dim) decoder reconstructing the masked 75%.
- Implemented patchify/unpatchify and random masking pipeline for 224×224 images into 196 tokens of 16×16 patches. Trained with Masked MSE Loss on TinyImageNet.

Hospital Management System — *C#, Oracle* 

- Built modules for patient registration, scheduling, and staff management with role-based access.
- Used parameterized SQL and stored procedures for secure and efficient database operations.

SKILLS

Languages Python, SQL, C++,GO, C#

Tools/Tech Ubuntu, Oracle Database, Git, Docker, PyTorch, Kaggle, FastAPI, FIGMA

Design Patterns Singleton, Adapter, Observer, MVC